

AlpaTrade — PDT-safe min-hold swing (buy_the_dip, Mag-7)

Realistic engine (engine.backtest: daily bars, next-open fills + 5bps slippage, N-1 Sharpe, no look-ahead) · basket AAPL, MSFT, GOOGL, AMZN, META, TSLA, NVDA · \$10,000 · 2025-07-20 → 2026-07-20.

Rule: buy the dip, hold a **minimum of 3 trading days** (never a same-day round trip → PDT-safe), then exit **intraday** once take-profit or stop is hit (intraday), with a max hold. Implemented as a new min_hold_days gate on the methodology strategy/engine.

Results (realistic fills)

Config (all min- hold 3d)	Dip / TP / SL / max- hold	Return	Annualised	Sharpe	Profit factor	Trades
tight-3d	3% / 1.5% / 3% / 10d	+0.99%	+1.00%	0.19	1.04	268
shallow- 3d	3% / 3% / 5% / 10d	+1.74%	+1.75%	0.29	1.07	210
mid-3d	5% / 5% / 8% / 15d	+4.28%	+4.32%	0.68	1.29	110
deep-3d	7% / 6% / 10% / 20d	+5.25%	+5.29%	0.88	1.45	69
— 1-day scalp (prior)	3% / 1.5% / 0.5% / 1d	-2.56%	-2.58%	-0.47	0.93	578
Buy & hold Mag-7	—	+21.31%	+21.50%	1.02	—	—

What this shows

- **The min-hold swing turns a loser into a real, positive-edge strategy.** Every 3-day-min config is profitable with **profit factor > 1 and positive Sharpe**, whereas the 1-day scalp (min-hold 0) *lost* money (PF 0.93,

Sharpe -0.47). Holding through the first few days — and only exiting intraday once the move actually materialises — is a genuine edge; scalping tiny same-week moves was just noise the costs ate.

- **Deeper dips + patience win.** deep-3d (buy 7% dips, 6% target, hold up to 20 days) is best — **PF 1.45, Sharpe 0.88, just 69 trades**. Fewer, higher-conviction trades beat frequent shallow ones.
- **But it still trails buy-and-hold (+21.3%) in this bull year.** With the Mag-7 trending up, sitting in cash between dips costs you the trend. The swing's advantage is *low exposure and low drawdown*, not out-gunning a raging bull.

Honest read

- This is now a **realistic** result (next-open fills, slippage, no intraday-scalp fantasy). A ~5% annual return with PF 1.45 is a modest but real edge — plausible, unlike the earlier +18%/+419% artifacts.
- In a **flat or choppy** market (where buy-and-hold stalls), a positive-PF dip-swing that's mostly in cash is exactly the kind of thing that adds value; in a **strong uptrend** it under-earns. Regime matters.
- Next: (a) test these configs in **down/sideways** windows to see where the swing beats buy-and-hold; (b) or promote deep-3d into the **paper-trading autonomy loop** and let real paper fills adjudicate.

Hypothetical; not financial advice. Regulatory fees excluded (execution friction only). All runs persist to the DB and to dated artifact folders under `backtest-results/`.